



Darshan UNIVERSITY

Python Programming - 2301CS404

Lab - 5

Roll No.:310

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1) Character Category Counter

Problem: Count uppercase, lowercase, digits, and special characters in a string.

Sample Inputs & Outputs:

- Input: Python3@Lab → Uppercase: 2, Lowercase: 7, Digits: 1, Special: 1
- Input: HELL0123 → Uppercase: 5, Lowercase: 0, Digits: 3, Special: 0
- Input: hi@#9 → Uppercase: 0, Lowercase: 2, Digits: 1, Special: 2

```
In [4]: s1 = input("Enter the string:")
uc=0;lc=0;dc=0;sc=0
for i in s1:
    if(i.isupper()):
        uc=uc+1
    elif(i.islower()):
        lc=lc+1
    elif(i.isdigit()):
        dc=dc+1
    else:
        sc=sc+1
print(f"Uppercase:{uc},Lowercase:{lc},Digits:{dc},Special:{sc}")
```

Enter the String: Python3@Lab
Uppercase:2,Lowercase:7,Digits:1,Special:1

2) Reverse Each Word

- Input: Python is Easy → nohtyP si ysaE
- Input: Hello World → olleH dlroW
- Input: Learn Python Fast → nraeL nohtyP tsaF

```
In [7]: s1 = input("Enter the string:")
sep = s1.split(" ")
rev=""
for word in sep:
    rev += word[::-1] + " "
rev
```

Enter the string: Python is Easy

Out[7]: 'nohtyP si ysaE '

3) Case Pattern Identifier

- PYTHON → Uppercase
- python → Lowercase
- Python Programming → Title Case

```
In [8]: s1 = input("Enter the string:")
if(s1.isupper()) :
    print("Uppercase")
elif(s1.islower()) :
    print("Lowercase")
elif(s1.istitle()) :
    print("Title Case")
```

Enter the string: PYTHON

Uppercase

4) First and Last Occurrence Finder

- programming , g → First:3 Last:10
- banana , a → First:1 Last:5
- hello , z → Not found

```
In [10]: s1 = input("Enter the string:")
ch = input("Enter the word to find:")
first = s1.find(ch)
last = s1.rfind(ch)
print(f"First:{first} Last:{last}")
```

Enter the string: programming

Enter the word to find: g

First:3 Last:10

5) Word Frequency Counter (ignore the case)

- Python is easy and Python is powerful → python: 2
- Java is popular but java is verbose → java: 2
- C is fast → c:1

```
In [13]: s1 = input("Enter the string:").lower()
ch = input("Enter the word to count:").lower()
count = s1.count(ch)
print(f"{ch}:{count}")
```

Enter the string: Python is easy and Python is powerful
 Enter the word to count: Python
 python:2

6) Remove Extra Spaces

- Python is fun → Python is fun
- Hello World → Hello World
- NoExtraSpace → NoExtraSpace

```
In [20]: s1 = input("Enter the string:")
s1.strip()
ans = " "
sep = s1.split()
ans+=" ".join(sep)
print(ans)
```

Enter the string: Python is fun
 Python is fun

7) Prefix and Suffix Removal

- unhappy.txt → happy
- pretest.py → test
- unwanted.doc → wanted

```
In [22]: s1 = input("Enter the string:")
prefix = input("Enter the prefix:")
suffix = input("Enter the suffix:")
s1 = s1.removeprefix(prefix)
s1 = s1.removesuffix(suffix)
print(s1)
```

Enter the string: unhappy.txt
 Enter the prefix: un
 Enter the suffix: .txt
 happy

8) Replace Vowels

- Programming → Pr*gr*mm*ng
- Education → *d*c*t**n
- Sky → Sky

```
In [28]: s1 = input("Enter the string:")
vowel=['A','a','E','e','I','i','O','o','U','u']
ans=""
for i in range(len(s1)) :
    ch=s1[i]
    if(ch in vowel) :
        s1 = s1.replace(ch, "*")
print(s1)
```

Enter the string: Programming
 Pr*gr*mm*ng

9) String Compression

- aaabbccccc → a3b2c4
- xxxyyyz → x3y3z1
- abcd → a1b1c1d1

```
In [40]: s1 = input("Enter the string:")
count=1
ans=""
for i in range(len(s1)-1) :
    if(s1[i]==s1[i+1]) :
        count+=1
    else :
        ans+= s1[i] + str(count)
        count = 1
ans += s1[-1] + str(count)
print(ans)
```

Enter the string: aaabbccccc
a3b2c4

10) Toggle Case (without using str.swapcase())

- PyThOn → pYtHoN
- HELlo → heLLo
- Python → pYTHON

```
In [41]: s1 = input("Enter the string:")
for i in s1:
    if(s1.isalpha()) :
        if(i.islower()) :
            s1 = s1.replace(i,i.upper())
        else :
            s1 = s1.replace(i,i.lower())
print(s1)
```

Enter the string: PyThOn
pYtHoN

11) Username Validation

valid username rules:

1. only alphabets and digits
2. does not start with a digit
3. length ≥ 6

- User123 → Valid
- 123User → Invalid
- Us@12 → Invalid

```
In [46]: s1 = input("Enter the string:")
if(i.isalpha() and len(s1)>=6 and not s1[0].isdigit()) :
    print("Valid")
```

```
else :
    print("Invalid")
```

Enter the string: User123
Valid

12) Palindrome Checker (ignore the case)

- Madam → Palindrome
- Level → Palindrome
- Python → Not Palindrome

```
In [52]: s1 = input("Enter the string:").lower()
if(s1==s1[::-1]):
    print("Palindrome")
else :
    print("Not Palindrome")
```

Enter the string: Madam
Palindrome

13) Longest Word Finder

- Python programming is interesting → programming
- I love coding → coding
- Data structures and algorithms → structures

```
In [54]: s1 = input("Enter the string:")
sep = s1.split(" ")
high_len = 0
for word in sep :
    if(len(word)>high_len) :
        high_len = len(word)
        hword = word
print(hword)
```

Enter the string: Python programming is interesting
programming

14) Case Conversion

- python programming
 - Upper: PYTHON PROGRAMMING
 - Lower: python programming
 - Title: Python Programming
 - Capitalize: Python programming

```
In [55]: s1 = input("Enter the string:")
print(f"Upper:{s1.upper()}")
print(f"Lower:{s1.lower()}")
print(f"Title:{s1.title()}")
print(f"Capitalize:{s1.capitalize()}")
```

Enter the string: python programming

Upper:PYTHON PROGRAMMING
Lower:python programming
Title:Python Programming
Capitalize:Python programming

15) Custom Split

- apple,banana,grape
- 10|20|30
- A-B-C-D

```
In [3]: s1 = input("Enter the string:")  
sep = input("Enter the sep:")  
ans = s1.split(sep)  
ans
```

```
Out[3]: ['apple', 'banana', 'grape']
```